Evangelion Japanese Font Metric for LuaTEX

https://github.com/RadioNoiseE/Evangelion-JFM

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Abstract

This documentation is going to introduce Evangelion Japanese Font Metric (hereinafter referred to as "Eva-JFM"), a Japanese Font Metric for typesetting high quality Chinese and Japanese documents. It can be used with Traditional Chinese, Simplified Chinese and Japanese fonts for both vertically and horizontally typesetted texts. It aims to provide a font metric which makes full use of the `priority` feature (provided by LuaTEX-ja), bases on the standard [1], and supports some advanced (a.k.a., rarely-used) features. The documentation is now written in both Chinese and English.

1 Background Information and a Rough Introduction

TEX is a powerful typesetting system "intended for the creation of beautiful books", it has full support for typesetting English based texts. However, its support for CJ text is limited[1]. For handling CJ texts in TEX, both macro extensions (i.e., CJK) and engine extensions were developed. One of the most influential one is (the) pTEX (series).

pTEX uses a virtual font scheme, by mapping TrueType or OpenType fonts using `TFM/VF` files. It doesn't support font configuration through macros, and has no support for PDF format output. Its advantage is the proven ability for dealing with traditional Japanese typographic layout requirements.

pdfTEX is a TEX engine extension which can directly output PDF files (just as its name). But it has limited support to Unicode as well as modern font formats (TrueType and OpenType vector font formats).

LuaTEX is based on pdfTEX. The inclusion of Lua enables it to support Unicode with the reader module, and modern fonts by using fontloader. Its macro based font setup feature is provided by luaotfload.

LuaTEX-ja can be seen as a porting of pTEX and LuaTEX. It's a macro package for typesetting high quality Japanese documents when using LuaTEX. LuaTEX supports font configuring by macros, therefore there's no need to keep pTEX's `VF` file. But for advanced features it left and extended[2] the so-called JFM file.

This document describes Eva-JFM, an advanced JFM file. By using LuaTEX's callback, it embeds features (maybe) needed in CJ text typesetting in `Eva-JFM.lua`. The features supported now are "Traditional Chinese", "Simplified Chinese", "Japanese", "Vertical Typesetting", "Linegap Punctuations", "Hanging Punctuations", "Extended Font", and "Non Standard".

2 Installation and Local Configurations

The sourcefiles are hosted on Github while it's also uploaded to CTAN. Users can simply use

```
tlmgr install evangelion-jfm
```

(or maybe using other package managers) to install. (But note that the CTAN branch is not always updated.) Developers can also use

[1] No Unicode, CJ's large charset, different writing systems, etc.

[2] The `priority` feature and some imaginary characters as well.

```
mkdir Evangelion-JFM [&&] cd Evangelion-JFM
git clone https://github.com/RadioNoiseE/Evangelion-JFM
```

to extract the latest version, the move it to the `TEXMF` directory, for instance

```
~/Library/texlive/2023/texmf-dist/tex/luatex/eva-jfm
```

If your TEX distribution requires

```
mktexlsr
```

to update the `Ls-R` database, make it so.

Eva-JFM doesn't requires any local configuration in most cases, but if you have some special requirements, have a look at section 5.3.

3 Useing

The above is an example of typesetting vertical text using Traditional Chinese fonts

```
\usepackage{luatexja-fontspec, luatexja-adjust}
\setmainjfont{Source Han Serif TC}[Language = Chinese Traditional, TateFeatures = {JFM = eva/{vert, trad
    , nstd}}]
\ltjenableadjust[priority = true]
```

(and be aware that you need to load a document class which supports vertical typesetting or use `\tate` command. LuaTEX-ja's JFM syntax is the above

```
jfm = ⟨JFM name⟩/{⟨JFM features⟩}
```

while under LATEX the most common case while using `\setmainjfont` is most likely

```
\setmainjfont{⟨font name⟩}[Language = ⟨language name⟩, ⟨dir⟩ = {JFM = ⟨JFM name⟩/{⟨JFM features⟩}}]
```

Option ⟨*font name*⟩ is the font (that you'd like to specify as the main font for your document)'s name. When using Japanese fonts, simply ignore the ⟨*language name*⟩ since LuaTEX-ja will automatically fill it for you. In this case, filling `Chinese Traditional` for Traditional Chinese fonts and `Chinese Simplified` for Simplified Chinese fonts is necessary[3]. ⟨*dir*⟩ should be `TateFeatures` when typeset vertically and `YokoFeatures` for typeset horizontally accordingly. The JFM's name is specified by the ⟨*JFM name*⟩ option[4]. Finally, for the ⟨*JFM features*⟩ key, fill in the JFM features. They are described in section 4.

For advanced users, it's also recommanded to use the following

```
\def\ltj@stdyokojfm{eva/{⟨JFM features⟩}}
```

or with the NFSS.

To set up JFM in other cases, please refer to the LuaTEX-ja document [2].

4 Supported Features

This section is going to give you a glance at all the features embedded in Eva-JFM. They are divided into 5 groups, they are Language Features, Direction Features, Extended Features, English Features and Dark Features respectively.

[3] Without this, your output may result in wrong details, for instance wrong punctuation shape & direction.

[4] LuaTEX-ja searchs for a JFM file by the rule `jfm-`⟨*JFM name*⟩`.lua`.

4.1 Language Features

You should specify one and only one feature from this section, or your TEX is going to complain about it.

`jp` ⟶ (*JaPanese*)
: Japanese font feature. When using Japanese fonts, you are required to specify this. It's very difference from Traditional Chinese and Simplified Chinese feature, namely the glue inserted after Question Mark and Exclamation Mark, and some punctuation mark's position when typeset vertically. It affects the feature `lgp`, as well as the internal grouping.

`trad` ⟶ (*TRADitional chinese*)
: Traditional Chinese feature. You should specify this when you are typesetting using Traditional Chinese fonts. The differences from the other two is because of its middle-placed punctuations. Hence the glues inserted next to it, the line-end adjust, as well as some kernings between punctuations are special.

`smpl` ⟶ (*SiMPLified chinese*)
: Simplified Chinese feature, for Simplified Chinese fonts. All the punctuations are laid down and placed aside. Therefore its position is treated with care. Eva-JFM also takes some rare conditions into consideration. Note that the *aki* after Question Mark and Exclamation Mark is different from that of the Japanese font feature.

4.2 Direction Features

Features in this section is compatible with all the other features.

`vert` ⟶ (*VERTical writing*)
: Vertical Typesetting feature. It affects kerning, internal grouping, etc. You should specify this when typeseting vertically.

4.3 Extended Features

Except the feature `hgp` doesn't rely on feature `vert`, all the other features need `vert` to work (since they should only be needed in vertical texts).

`extd` ⟶ (*EXTenDed font*)
: Extended font features. The dafault ratio is $x : y = 100 : 80$ while x is the width and y is the height. You can customize it using `extd=`⟨*ratio*⟩ (the dafault ⟨*ratio*⟩ is 1.25). It should be used with `extend` (luaotfload) or `FakeStretch` (fontspec).

`lgp` ⟶ (*LineGap Punctuations*)
: The linegap punctuations feature. This hangs some punctuations into the linegap. Some difference occurs when it's used with the `jp` feature. For more information see section 5.

`hgp` ⟶ (*HanGing Punctuations*)
: Hanging punctuation feature which "hangs" some punctuation at line-end (allowing them to stick out a bit). Traditional Chinese fonts doesn't support this feature because the result is somewhat (rather) wierd.

4.4 English Features

You need to set the JAchar range using `\ltjsetparameter` before using features in this section. It's also recommended to use with the corresponding OpenType features.

`hwid` ⟶ (*Half WIDth*)

Half width English characters feature. This will place each alphabets into a box which width is exactly 0.5 times the CJ character's width. It's worth noting that it will not stretch or shrink the glyph, it only adjusts the spacing. Hence if the OpenType feature `hwid` is not set, English characters will simply overlap. All the kernings and italic corrections will also be lost (this may be fixed in the future versions), and will ignore the parameter `xkanjiskip`. Please use with care.

`fwid` ⟶ (*Full WIDth*)

Full width English characters feature. It's similar from feature `hwid` above except that the spacing will be stretched out on the contrary.

4.5 Dark Features

Before using the following features, please make sure that you have carefully read the descriptions.

`nstd` ⟶ (*Non STandarD*)

This one ignores the standard priority rules for punctuation kerning. While Japanese text layout requirement [1] suggests that the priority for the period should be higher than the comma (which means the period is easier to stretch), this makes the comma's priority higher than the preiod's. Only works when luatexja-adjust's priority feature is enabled (set to `true`).

5 Linegap Punctuation Feature

Here more detailed information about linegap punctuations are provided, as well as the issues may occur and the possible solution.

5.1 About "Hanging"

Linegap punctuations can be seen in Chinese ancient books, it's a combination of the punctuations marks and the traditional vertical typesetting method.

Only periods and commas should be hanged but Eva-JFM hangs three more punctuations in addition. Japanese font is different in this aspect however, since the direction of colon and semicolon makes it impossible to be hanged.

They are all hanged to the lower right of the glyph. See the next subsection for more details.

5.2 Hanging Position

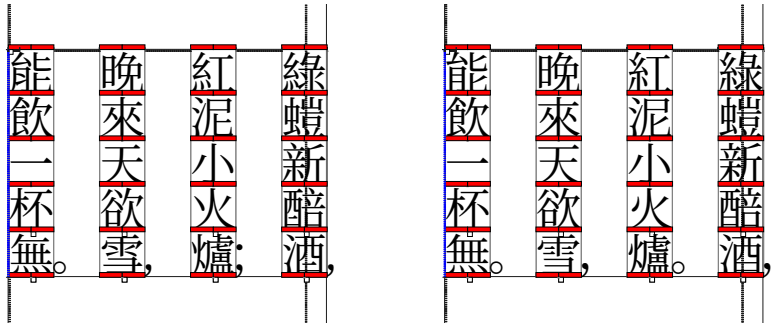

Figure 1: The linegap punctuations feature

The position of these hanged punctuations is decided according to the following rules as shown in the figure 1. For customizing, see subsection 5.3. The rules which occurs more early have the higher priorities.

- The style of the three fonts are unified;
- The position of the similar elements in different punctuations should be the same;
- The glyph of the punctuations should touch the *kanji*'s boundary;
- Different punctuations' position can vary considering their glyphs' shapes, sizes, design respectively.

5.3 User Configs

This feature is designed for the Source Han font series (思源系列). Due to different fonts' different punctuation marks, the output may be wrong (overlap, not aligned, etc). Also you may prefer your own settings. Therefore, two methods of customizing the positions of hanged punctuations is provided here.

5.4 Changing Parameters

In Eva-JFM, the tables which contains the parameters for the positions of these hanged punctuations is

```
[101,2] ==> [1]; [201,2] ==> [2]; [301,2] ==> [3].
```

Kindly modify `left` (dir right) and `down` (dir down) until the output is fine. You can also refer to the last section (*Implementing*).

5.5 Using Extra Font

Extracting the glyphs for punctuation marks and package them into a new font (you can use programs like *fontforge*) and use them for hanging punctuations later is the second solution. You can also load another font just for its punctuations (but loading a CJ font into TEX's memory has an expensive cost).

After installing that font, you can use the `AltFont` key provided by LuaTEX-ja to replace the punctuations. The actual code is shown above.

```
\setmainjfont[
    Language = ⟨language⟩,
    TateFeatures = {
        JFM = eva/{vert, lgp, ⟨language⟩},
        AltFont = {
            {Range = "⟨utf-8 code⟩, Font = ⟨symbol font⟩}
        }
    }
]{⟨main font⟩}
```

One of `Japanese`, `Chinese Traditional` or `Chinese Simplified` should be filled in the first ⟨*language*⟩ option, the other one is for the corresponding JFM features. ⟨*utf-8 code*⟩ selects the punctuations you'd like to replace with the "punctuation font"[5]. Finally, it's obvious that the ⟨*symbol font*⟩ and the ⟨*main font*⟩ options are for the "punctuation font" and the main font.

It's also recommended for the developers to use the NFSS with

```
\DeclareAlternateKanjiFont{⟨base encoding⟩}{⟨base family⟩}{⟨base series⟩}{⟨base shape⟩}{⟨alt encoding⟩}{⟨alt
    family⟩}{⟨alt series⟩}{⟨alt shape⟩}{⟨range⟩}
```

Option ⟨*base*⟩ and ⟨*alt*⟩ stands for main font and "punctuation font".

Refer to the LuaTEX-ja document [2] for more detailed syntax and usage as well as some examples.

[5] You can search `https://www.unicode.org/charts/unihanrsindex.html` for their unicodes representations.

6 Inspiration

Eva-JFM's internal grouping is inspired by `min10.tfm` [5], while its `priority` feature's data partly comes from Noriyuki Abe's `jlreq.lua` [6].

References

[1] W3C Japanese Layout Task Force (ed). Requirements for Japanese Text Layout (W3C Working Group Note), 2022, 2023. https://www.w3.org/TR/jlreq/.

[2] LuaTEX-ja プロジェクトチーム. LuaTEX-ja パッケージ, 2022, 2023.

[3] The Unicode Consortium. The Unicode Standard Version 15.0 - Core Specification, 2022.

[4] Victor Eijkhout. TEX by Topic, A TEXnician's Reference, Addison-Wesley, 1992.

[5] 乙部厳己. min10 フォントについて. http://argent.shinshu-u.ac.jp/~otobe/tex/files/min10.pdf.

[6] Noriyuki Abe. Jlreq Document Class, 2022. https://github.com/abenori/jlreq.

Implementation

The above is the implementation of this font metric. Can be used for reference.

```
---- Evangelion Japanese Font Metric for LuaTeX
---- Current Version: 1.0.2 (b)
---- Dev URL: https://github.com/RadioNoiseE/Evangelion-JFM
---- © Copyright 2023, RadioNoiseE


-- 初始化
local lang_jp, lang_tc, lang_sc, dir_vt, font_extd, punc_lg, punc_hg, std_nil, al_hw, al_fw

if luatexja.jfont.jfm_feature then
    lang_jp = luatexja.jfont.jfm_feature.jp
    lang_tc = luatexja.jfont.jfm_feature.trad
    lang_sc = luatexja.jfont.jfm_feature.smpl
    dir_vt = luatexja.jfont.jfm_feature.vert
    font_extd = luatexja.jfont.jfm_feature.extd
    punc_lg = luatexja.jfont.jfm_feature.lgp
    punc_hg = luatexja.jfont.jfm_feature.hgp
    std_nil = luatexja.jfont.jfm_feature.nstd
    al_hw = luatexja.jfont.jfm_feature.hwid
    al_fw = luatexja.jfont.jfm_feature.fwid
end

-- 預處理及容錯
if font_extd == true and dir_vt == false then
    tex.error('JFM feature "extd" only works with feature "vert".\n' ..
              'For now I\'ll ignore it.')
end

if punc_lg == true and dir_vt == false then
    tex.error('JFM feature "lgp" only works with feature "vert".\n' ..
              'For now I\'ll ignore it.')
```

```lua
end

if al_hw == true and al_fw == true then
    tex.error('JFM feature "hwid" cannot be used with "fwid".')
end

if not ((lang_jp and not (lang_tc or lang_sc)) or
    (lang_tc and not (lang_jp or lang_sc)) or
    (lang_sc and not (lang_jp or lang_tc))) then
    tex.error('Specify one and only one feature from three language specific features\n' ..
              '"jp", "trad" or "smpl"\n' ..
              'is required.\n' ..
              'For now I\'ll use "lang_jp" for japanese by default.')
end

-- 壓縮比例設定
if font_extd == true then
    local extd_ratio = (type(luatexja.jfont.jfm_feature and font_extd) == 'string') and tonumber(
     font_extd) or 1.25
end

-- 行間標點字間距補足
local lgp_kanjiskip = {kanjiskip_natural = 0, kanjiskip_stretch = 1, kanjiskip_shrink = 1}

-- 定義函數宏
local function logic_anif(f1, f2, r1, r2)
     local rta = f1 and (f2 and r1) or r2
     return rta
end

local function logic_if(f1, r1, r2)
     local rti = f1 and r1 or r2
     return rti
end

local function context_height()
    local rth = dir_vt and (font_extd and extd_ratio/2 or 0.5) or 0.88
    return rth
end

local function context_depth()
    local rtd = dir_vt and (font_extd and extd_ratio/2 or 0.5) or 0.12
    return rtd
end

-- 主體
local eva = {
    version = 3,
    dir = logic_if(dir_vt, 'tate', 'yoko'),
    zw = 1,
    zh = logic_anif(dir_vt, font_extd, extd_ratio, 1),
    kanjiskip = {0, 0.25, 0},
    xkanjiskip = {0.25, 0.125, 0.125},

    [0] = { -- 缺省類
        width = 1,
```

```
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = 'middle',
        glue = {
            [1] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = logic_if(std_nil, {-1, 0}, {-1, -2})}, {priority = logic_if(std_nil, {-1, 0}, {-1, -2})}),
            [2] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = logic_if(std_nil, {-1, -2}, {-1, 0})}, {priority = logic_if(std_nil, {-1, -2}, {-1, 0})}),
            [3] = logic_if(dir_vt, {priority = {0, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = {-1, -1}}, {priority = {-1, -1}})),
            [7] = {0.5, 0, 0.25, ratio = 1, priority = {-1, -2}},
            [9] = {0.25, 0, 0.125, ratio = 1, priority = {-1, -1}}
        },
        round_threshold = 0.01
    },

    [1] = { -- 読点類
        chars = logic_anif(dir_vt, punc_lg, {}, {'、', ','}),
        width = 0.5,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = logic_if(lang_tc, 'middle', 'left'),
        glue = {
            [0] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {-1, 0}, {-1, -2})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {-1, 0}, {-1, -2})}),
            [1] = logic_if(lang_tc, {0.5, 0, 0.25}, {0.5, 0, 0.25}),
            [2] = logic_if(lang_tc, {0.5, 0, 0.25}, {0.5, 0, 0.25}),
            [3] = logic_if(dir_vt, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}, {0.5, 0, 0.25, priority = {0, -1}}), logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}})),
            [4] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {0, 0}, {0, -2})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {0, 0}, {0, -2})}),
            [5] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {0, 0}, {0, -2})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {0, 0}, {0, -2})}),
            [6] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {-1, 0}, {-1, -2})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {-1, 0}, {-1, -2})}),
            [7] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {-1, 0}, {-1, -2})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {-1, 0}, {-1, -2})}),
            [8] = logic_if(lang_tc, {0.25, 0, 0.125}, {}),
            [9] = logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.75, 0, 0.25, ratio = 1/3, priority = {0, -1}})
        },
        end_adjust = logic_if(lang_tc, {0.25, 0}, logic_if(punc_hg, {-0.5, 0}, {0, 0}))
    },

    [101] = { -- 読点類（行間a）
        chars = logic_anif(dir_vt, punc_lg, {'、'}, {}),
        width = 0,
        height = context_height(),
        depth = context_depth(),
```

```
        italic = 0,
        left = 0.38,
        down = -0.34,
        align = 'left',
        glue = {
            [0] = lgp_kanjiskip
        }
    },

    [102] = { -- 読点類（行間b）
        chars = logic_anif(dir_vt, punc_lg, {','}, {}),
        width = 0,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = logic_if(lang_tc, 0.62, 0.40),
        down = logic_if(lang_tc, -0.58, -0.26),
        align = 'left',
        glue = {
            [0] = lgp_kanjiskip
        }
    },

    [2] = { -- 句點類
        chars = logic_anif(dir_vt, punc_lg, {}, {'.', '。'}),
        width = 0.5,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = logic_if(lang_tc, 'middle', 'left'),
        glue = {
            [0] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {-1, -2}, {-1, 0})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {-1, -2}, {-1, 0})}),
            [1] = logic_if(lang_tc, {0.5, 0, 0.25}, {0.5, 0, 0.25, ratio = 0}),
            [2] = logic_if(lang_tc, {0.5, 0, 0.25}, {0.5, 0, 0.25, ratio = 0}),
            [3] = logic_if(dir_vt, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}, {0.5, 0, 0.25, priority = {0, -1}}), logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}})),
            [4] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {0, -2}, {0, 0})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {0, -2}, {0, 0})}),
            [5] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {0, -2}, {0, 0})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {0, -2}, {0, 0})}),
            [6] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {-1, -2}, {-1, 0})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {-1, -2}, {-1, 0})}),
            [7] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = logic_if(std_nil, {-1, -2}, {-1, 0})}, {0.5, 0, 0.25, ratio = 0, priority = logic_if(std_nil, {-1, -2}, {-1, 0})}),
            [8] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0}, {}),
            [9] = logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.75, 0, 0.25, ratio = 1/3, priority = {0, -1}})
        },
        end_adjust = logic_if(lang_tc, {0.25, 0}, logic_if(punc_hg, {-0.5, 0}, {0, 0}))
    },

    [201] = { -- 句點類（行間a）
```

```
        chars = logic_anif(dir_vt, punc_lg, {'.'}, {}),
        width = 0,
        height = context_height(),
        depth = context_height(),
        italic = 0,
        left = logic_if(lang_tc, 0.68, 0.34),
        down = logic_if(lang_tc, -0.58, -0.28),
        align = 'left',
        glue = {
            [0] = lgp_kanjiskip
        }
    },

    [202] = { -- 句點類（行間b）
        chars = logic_anif(dir_vt, punc_lg, {'。'}, {}),
        width = 0,
        height = context_height(),
        depth = context_height(),
        italic = 0,
        left = 0.42,
        down = -0.35,
        align = 'left',
        glue = {
            [0] = lgp_kanjiskip
        }
    },

    [3] = { -- 兩點類
        chars = logic_if(lang_jp, {}, (logic_anif(dir_vt, punc_lg, {}, {'：', '；'}))),
        width = logic_if(dir_vt, 1, 0.5),
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = logic_if(lang_tc, 'middle', 'left'),
        glue = {
            [0] = logic_if(dir_vt, {priority = {-1, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {-1, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {-1, -1}})),
            [1] = logic_if(dir_vt, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}, {priority = {0, -1}}), logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}})),
            [2] = logic_if(dir_vt, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}, {priority = {0, -1}}), logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}})),
            [3] = logic_if(dir_vt, {priority = {0, -1}}, logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}})),
            [4] = logic_if(dir_vt, {priority = {0, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}})),
            [5] = logic_if(dir_vt, {priority = {0, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}})),
            [6] = logic_if(dir_vt, {priority = {-1, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {-1, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {-1, -1}})),
            [7] = logic_if(dir_vt, {priority = {-1, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {-1, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {-1, -1}})),
```

```
            [8] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}, {0.5, 0, 0.25, ratio = 0, priority = {0, -1}}),
            [9] = logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.75, 0, 0.25, ratio = 1/3, priority = {0, -1}})
        }
    },

    [301] = { -- 兩點類（行間a）
        chars = logic_if(lang_jp, {}, logic_anif(dir_vt, punc_lg, {'：'}, {})),
        width = 0,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = logic_if(lang_tc, 0.94, 0.72),
        down = logic_if(lang_tc, -0.58, -0.34),
        align = 'left',
        glue = {
            [0] = lgp_kanjiskip
        }
    },

    [302] = { -- 兩點類（行間b）
        chars = logic_if(lang_jp, {}, logic_anif(dir_vt, punc_lg, {'；'}, {})),
        width = 0,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = logic_if(lang_tc, 0.96, 0.78),
        down = logic_if(lang_tc, -0.58, -0.34),
        align = 'left',
        glue = {
            [0] = lgp_kanjiskip
        }
    },

    [4] = { -- 小書きの仮名類
        chars = {
            'ぁ', 'ぃ', 'ぅ', 'ぇ', 'ぉ', 'っ', 'ゃ', 'ゅ', 'ょ', 'ゎ', 'ゕ',
            'ゖ', 'ゝ', 'ゞ', 'ァ', 'ィ', 'ゥ', 'ェ', 'ォ', 'ッ', 'ャ', 'ュ',
            'ョ', 'ヮ', 'ヵ', 'ヶ', 'ヽ', 'ヾ', 'ク', 'シ', 'ス', 'ト', 'ヌ',
            'ハ', 'ヒ', 'フ', 'ヘ', 'ホ', 'ム', 'ラ', 'リ', 'ル', 'レ', 'ロ'
        },
        width = 1,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = 'middle',
        glue = {
            [1] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, {}),
            [2] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, {}),
            [3] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}, {priority = {0, -1}}),
            [7] = {0.5, 0, 0.25, ratio = 1, priority = {-1, -2}},
            [9] = {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}
```

```
        }
    },

    [5] = { -- 疑問感嘆類
        chars = {'！', '？', '‼ 🈖', '⁉ 🈖', '?!', '??'},
        width = logic_if(dir_vt, 1, logic_if(lang_sc, 0.5, 1)),
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = logic_if(dir_vt, 'middle', logic_if(lang_sc, 'left', 'middle')),
        glue = {
            [0] = logic_if(dir_vt, logic_if(lang_jp, {1, 0, 0.5, ratio = 0, priority = {-1, 0}}, {priority = {-1, 0}}), logic_if(lang_tc, {priority = {-1, 0}}, {0.5, 0, 0.25, ratio = 0, priority = {-1, 0}})),
            [1] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, logic_anif(not dir_vt, lang_sc, {0.5, 0, 0.25, ratio = 0}, {})),
            [2] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, logic_anif(not dir_vt, lang_sc, {0.5, 0, 0.25, ratio = 0}, {})),
            [3] = logic_if(dir_vt, {priority = {-1, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = {-1, -1}}, {0.75, 0, 0.25, ratio = 1/3, priority = {-1, -1}})),
            [4] = logic_if(dir_vt, logic_if(lang_jp, {1, 0, 0.5, ratio = 0}, {}), logic_if(lang_tc, {}, {0.5, 0, 0.25, ratio = 0})),
            [7] = {0.5, 0, 0.25, ratio = 1, priority = {-1, -2}},
            [8] = logic_anif(not dir_vt, lang_sc, {0.5, 0, 0.25, ratio = 0}, {}),
            [9] = logic_anif(not dir_vt, lang_sc, {0.75, 0, 0.25, ratio = 1/3, priority = {-1, -1}}, {0.25, 0, 0.125, ratio = 1, priority = {-1, -1}})
        }
    },

    [6] = { -- 分離禁止類
        chars = {'—', '—', '‥', '…', '…', '／', '⁄', '＼'},
        width = 1,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = 'middle',
        kern = {
            [6] = 0
        },
        glue = {
            [1] = logic_if(lang_tc, {0.25, 0, 0,125, ratio = 1}, {}),
            [2] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, {}),
            [3] = logic_if(dir_vt, {priority = {0, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}, {priority = {0, -1}})),
            [7] = {0.5, 0, 0.25, ratio = 1, priority = {-1, -2}},
            [9] = {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}
        }
    },

    [7] = { -- 開括號類
        chars = {'（', '〔', '［', '｛', '〈', '《', '「', '『', '【', '〖', '〘', '〚', '〝', '‘', '“'},
```

```
        width = 0.5,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = 'right',
        glue = {
            [1] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, {}),
            [2] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, {}),
            [3] = logic_if(dir_vt, {priority = {0, -1}}, logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}, {priority = {0, -1}})),
            [7] = {0, 0, 0, priority = {0, 0}, kanjiskip_shrink = 1},
            [9] = {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}
        }
    },

    [8] = { -- 閉括號類
        chars = {')', '〕', '］', '｝', '〉', '》', '」', '』', '】', '〕', '〗', '〙', '〛', '’', '”'},
        width = 0.5,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = 'left',
        glue = {
            [0] = {0.5, 0, 0.25, ratio = 0, priority = {-1, -2}},
            [1] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, {}),
            [2] = logic_if(lang_tc, {0.25, 0, 0.125, ratio = 1}, {}),
            [3] = logic_if(lang_sc, {priority = {0, -1}}, {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}),
            [4] = {0.5, 0, 0.25, ratio = 0, priority = {0, -2}},
            [5] = {0.5, 0, 0.25, ratio = 0, priority = {0, -2}},
            [6] = {0.5, 0, 0.25, ratio = 0, priroity = {-1, -2}},
            [7] = {0.5, 0, 0.25, ratio = 0, priority = {-1, -2}, kanjiskip_stretch = 1},
            [8] = {0, 0, 0, priority = {0, 0}, kanjiskip_shrink = 1},
            [9] = {0.25, 0, 0.125, ratio = 1, priority = {0, -1}}
        }
    },

    [9] = { -- 中點類
        chars = logic_if(lang_jp, {'・', '：', '；'}, {'・', '·'}),
        width = 0.5,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = 'middle',
        glue = {
            [0] = {0.25, 0, 0.125, ratio = 0, priority = {-1, -1}},
            [1] = logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}),
            [2] = logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}),
```

```
            [3] = logic_if(dir_vt, {0.25, 0, 0.125, ratio = 0, priority = {0, -1}}, logic_if(lang_tc, {0.5, 0, 0.25, priority = {0, -1}}, {0.25, 0, 0.125, priority = {0, -1}})),
            [4] = {0.25, 0, 0.125, ratio = 0, priority = {0, -1}},
            [5] = {0.25, 0, 0.125, ratio = 0, priority = {0, -1}},
            [6] = {0.25, 0, 0.125, ratio = 0, priority = {-1, -1}},
            [7] = {0.25, 0, 0.125, ratio = 0, priority = {-1, -1}},
            [8] = {0.25, 0, 0.125, ratio = 0, priority = {0, -1}},
            [9] = {0.5, 0, 0.25, priority = {0, -1}}
        }
    },

    [10] = { -- 西文
        chars = {},
        width = 0,
        height = context_height(),
        depth = context_depth(),
        italic = 0,
        left = 0,
        down = 0,
        align = 'middle',
        glue = {}
    },

    [11] = { -- 行頭
        chars = {'parbdd', 'boxbdd'},
        glue = {
            [7] = {0, 0, 0}
        }
    }
}

if al_hw == true or al_fw == true then
    eva[10].chars = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm',
                     'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z',
                     'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M',
                     'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z'}
    eva[10].glue = table.fastcopy(eva[0].glue)
    eva[10].glue[0] = {0.25, 0.125, 0.125, ratio = 0, priority = {0, -1}}
    eva[0].glue[10] = {0.25, 0.125, 0.125, ratio = 1, priority = {0, -1}}
    for _, catnum in ipairs({1, 2, 3, 5, 8, 9}) do
        eva[catnum].glue[10] = table.fastcopy(eva[catnum].glue[0])
    end
end

if al_hw == true and al_fw == false then
    eva[10].width = 0.5
end

if al_fw == false and al_fw == true then
    eva[10].width = 1
end

luatexja.jfont.define_jfm(eva)
```